

中秋猜燈謎

Mid-Autumn Lantern Riddle

Counting a “full house” with combinations

Participation exercise · please show your working

謎面 · The riddle

三人同顏一對和，
五張牌裡月婆娑。
花燈若問幾許亮？
請君細數莫猜錯。

Three of one face, and a matching pair — five cards under the Mid-Autumn moon. How many such lanterns can a full deck light?

Decode the riddle first. 「三人同顏」 means three cards of the *same number*. 「一對和」 means one *pair* (two cards of another number). Together this is a **full house**: three-of-a-kind + one pair.

Deck rules (read carefully). A standard deck has **52 cards**: **4 suits** (kinds), and in each suit the numbers run from **1 to 13**. So there are 13 ranks, and for each rank there are exactly 4 cards (one per suit). You form a hand of **5 cards**. Order does *not* matter — count with combinations $C(n, k)$, not permutations.

解謎任務 · Your tasks

1. **Pattern check.** In one sentence, state what hand the riddle is asking you to count (three of one rank + a pair of another rank).

2. **Build the count with combinations.** Write a product of combination symbols, then evaluate it, for the number of distinct full-house hands.

Suggested steps (you may rearrange, but show the reasoning):

- choose the rank for the three-of-a-kind;
- choose which 3 of the 4 suits for that rank;
- choose the rank for the pair (different from the first);
- choose which 2 of the 4 suits for that pair.

3. **(Optional bonus)** There are $C(52, 5)$ possible 5-card hands. Write the probability of a full house as a fraction (you need not fully simplify by hand if the numbers are large).

Name: _____ SID / Group: _____ Date: _____

Submit as instructed on Blackboard. This counts toward **course participation** (in-class exercises). 中秋快樂· Happy Mid-Autumn Festival.